

What is a System

The word System is derived from Greek word Systema, which means an organized relationship between any set of components to achieve some common cause or objective.

A system is “an orderly grouping of interdependent components linked together according to a plan to achieve a specific goal.”

Constraints of a System

A system must have three basic constraints –

- A system must have some **structure and behavior** which is designed to achieve a predefined objective.
- **Interconnectivity** and **interdependence** must exist among the system components.
- The **objectives of the organization** have a **higher priority** than the objectives of its subsystems.

For example, traffic management system, payroll system, automatic library system, human resources information system.

EXAMPLE: A business organization is a system with its components as:

Marketing, Manufacture, Sales, Research, Shipping, Accounting, Personnel, Administrative
All these components work together to create a profit.

Characteristics of a System

A system has the following properties –

Organization

Organization implies structure and order. It is the arrangement of components that helps to achieve predetermined objectives.

Interaction

It is defined by the manner in which the components operate with each other.

For example, in an organization, purchasing department must interact with production department and payroll with personnel department.

Interdependence

Interdependence means how the components of a system depend on one another. For proper functioning, the components are coordinated and linked together according to a specified plan. The output of one subsystem is the required by other subsystem as input.

Integration

Integration is concerned with how a system components are connected together. It means that the parts of the system work together within the system even if each part performs a unique function.

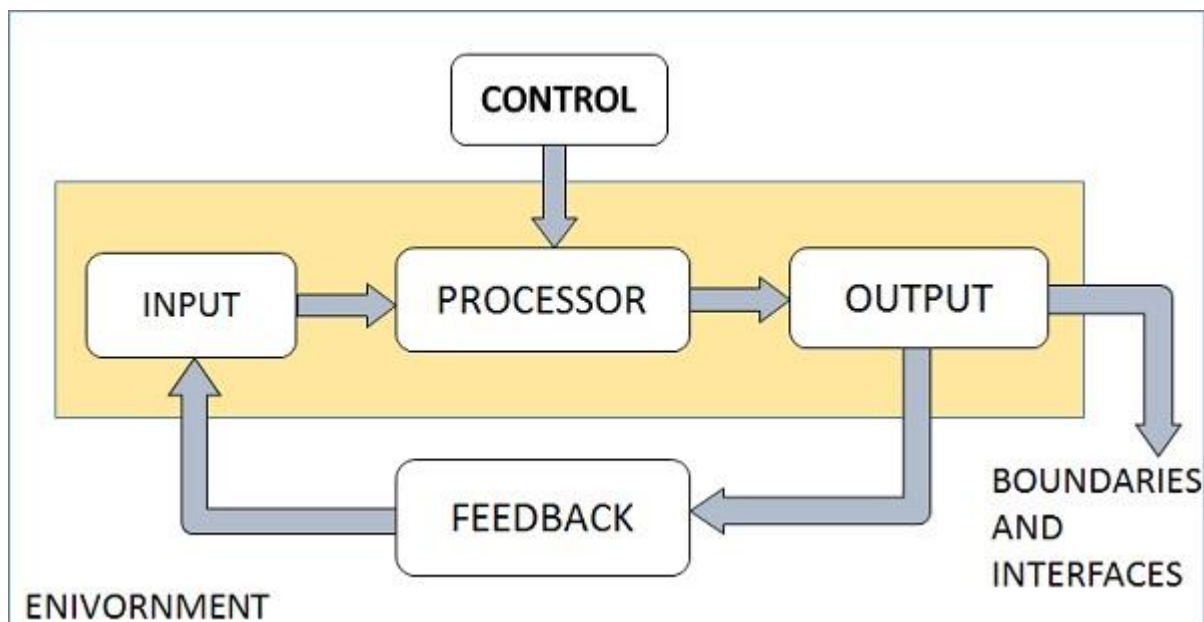
Central Objective

The objective of system must be central. It may be real or stated. It is not uncommon for an organization to state an objective and operate to achieve another.

The users must know the main objective of a computer application early in the analysis for a successful design and conversion

Elements of a System

The following diagram shows the elements of a system –



Outputs and Inputs

- The main aim of a system is to produce an output which is useful for its user.
- Inputs are the information that enters into the system for processing.
- Output is the outcome of processing.

Processor(s)

- The processor is the element of a system that involves the actual transformation of input into output.
- It is the operational component of a system. Processors may modify the input either totally or partially, depending on the output specification.
- As the output specifications change, so does the processing. In some cases, input is also modified to enable the processor for handling the transformation.

Control

- The control element guides the system.
- It is the decision-making subsystem that controls the pattern of activities governing input, processing, and output.
- The behavior of a computer System is controlled by the Operating System and software. In order to keep system in balance, what and how much input is needed is determined by Output Specifications.

Feedback

- Feedback provides the control in a dynamic system.
- Positive feedback is routine in nature that encourages the performance of the system.
- Negative feedback is informational in nature that provides the controller with information for action.

Environment

- The environment is the “super system” within which an organization operates.
- It is the source of external elements that strike on the system.
- It determines how a system must function. For example, vendors and competitors of organization’s environment, may provide constraints that affect the actual
- performance of the business.

Boundaries and Interface

- A system should be defined by its boundaries. Boundaries are the limits that identify its components, processes, and interrelationship when it interfaces with another system.
- Each system has boundaries that determine its sphere of influence and control.
- The knowledge of the boundaries of a given system is crucial in determining the nature of its interface with other systems for successful design.

Types of Systems

The systems can be divided into the following types –

Physical or Abstract Systems

- Physical systems are tangible entities. We can touch and feel them.
- Physical System may be static or dynamic in nature. For example, desks and chairs are the physical parts of computer center which are static. A programmed computer is a dynamic system in which programs, data, and applications can change according to the user's needs.
- Abstract systems are non-physical entities or conceptual that may be formulas, representation or model of a real system.
- For example, an educational system: Our social system.

Open or Closed Systems

- An open system must interact with its environment. It receives inputs from and delivers outputs to the outside of the system. For example, an information system which must adapt to the changing environmental conditions.
- A closed system does not interact with its environment. It is isolated from environmental influences. A completely closed system is rare in reality.

Adaptive and Non Adaptive System

- Adaptive System responds to the change in the environment in a way to improve their performance and to survive. For example, human beings, animals.
- Non Adaptive System is the system which does not respond to the environment. For example, machines.

Natural and Manufactured System

- Natural systems are created by the nature and there is no interference of human.
- . For example, Solar system, seasonal system.
- **Manufactured System** is the made by the human.
- It is also known as man made system
- For example, Rockets, dams, trains.

Deterministic or Probabilistic System

- Deterministic system operates in a predictable manner and the interaction between system components is known with certainty. For example, two molecules of hydrogen and one molecule of oxygen makes water.

- Probabilistic System shows uncertain behavior. The exact output is not known. For example, Weather forecasting, mail delivery.

Social, Human-Machine, Machine System

- Social System is made up of people. For example, social clubs, societies.
- In Human-Machine System, both human and machines are involved to perform a particular task. For example, Computer programming.
- Machine System is where human interference is neglected. All the tasks are performed by the machine. For example, an autonomous robot.
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SYSTEM APPROACH: -

A "system approach" refers to a method of problem-solving and analysis that focuses on understanding and addressing complex issues by considering the interactions and interdependencies within a larger system, rather than isolating individual components. This approach is widely used in various fields, including engineering, management, ecology, and more. It aims to enhance our understanding of the underlying dynamics of a system and how its components work together to achieve specific goals.

Key principles of a system approach include:

1. **Holistic Perspective:** Instead of viewing a problem in isolation, the system approach encourages looking at the bigger picture and understanding how various elements are interconnected and influence each other.
2. **Interdependence:** Recognizing that the components of a system are interdependent and that changes to one component can have ripple effects throughout the system. This emphasizes the importance of considering unintended consequences.
3. **Feedback Loops:** Understanding that systems often include feedback loops, where outputs from a component can affect inputs to the same or other components. Feedback loops can be reinforcing (positive) or balancing (negative).
4. **Boundaries and Context:** Defining the boundaries of the system being studied and considering its context within larger systems. This helps in defining the scope of analysis and understanding the external factors that impact the system.

5. **Emergent Properties:** Acknowledging that a system can exhibit properties and behaviors that are not directly attributable to its individual components but emerge from their interactions. These emergent properties can be difficult to predict based solely on the characteristics of the components.
6. **Multi-disciplinary Approach:** A system approach often involves collaboration among experts from different disciplines to ensure a comprehensive understanding of all relevant aspects of the system.
7. **Long-Term Perspective:** Considering the long-term consequences and sustainability of decisions and actions taken within the system. This involves understanding how short-term changes might impact the system's stability and functionality over time.
8. **Modeling and Simulation:** Creating models and simulations to represent the system's behavior and test various scenarios. This can help in understanding potential outcomes of different interventions or changes within the system.
9. **Continuous Learning and Adaptation:** Recognizing that systems are dynamic and subject to change, and therefore requiring ongoing monitoring, learning, and adaptation to maintain their functionality and relevance.

The system approach is particularly valuable when dealing with complex and interconnected challenges that cannot be adequately addressed through isolated solutions. It encourages a more comprehensive and informed decision-making process by taking into account the broader context and potential consequences of actions within a system.

INFORMATION SYSTEM: -

An information system (IS) refers to a coordinated collection of hardware, software, data, people, and procedures designed to gather, process, store, and disseminate information within an organization or to support specific business processes. Information systems play a critical role in modern business operations, decision-making, communication, and overall management. They help organizations streamline processes, improve efficiency, and make informed decisions based on data analysis.

CHARACTERSTICES OF INFORMATION SYSTEM:

1. **Data Input:** Information systems receive data from various sources, such as users, sensors, or other systems. This data can be in various formats, such as text, numbers, images, or audio.
 2. **Data Processing:** The collected data is processed, which involves organizing, analyzing, and transforming it into meaningful information through various algorithms and computations.
 3. **Data Storage:** Processed information is stored in databases or other data storage mechanisms to make it available for future use and retrieval.
 4. **Data Output:** The information system delivers the processed data in a format that is useful and understandable to users through reports, dashboards, or user interfaces.
 5. **Feedback Mechanism:** Information systems often have feedback loops, where the output or results of a process are used to modify and improve subsequent inputs or processes.
 6. **Interactivity:** Many information systems allow users to interact with the data, analyze it, and perform various actions to achieve their objectives.
 7. **Integration:** Information systems integrate data and processes from various sources and departments within an organization to create a unified and cohesive system.
 8. **Security:** Ensuring data security and privacy is a crucial aspect of information systems to protect sensitive information from unauthorized access or misuse.
 9. **Scalability:** Information systems should be able to handle increasing amounts of data and users as an organization grows.
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Role of information in decision making

Information plays a crucial role in the decision-making process. It serves as the foundation for informed decisions. Also, it helps individuals and organizations make choices that are based on accurate and relevant data. The following are some of the key ways in which information contributes to decision making:

1. **Problem Solving:** Information helps identify problems and allows decision makers to evaluate alternative solutions and choose the best course of action.
 2. **Evidence-based Decision Making:** Information provides the evidence needed to make decisions that are based on data and facts, rather than opinions or assumptions.
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3. Risk Assessment: Information helps decision makers assess risks and make decisions that minimize potential harm and maximize benefits.

4. Improved Accuracy: Information helps decision makers make more accurate decisions by providing them with a complete and accurate picture of a situation

TYPES OF INFORMATION SYSTEM: -

1. Transaction Processing System (TPS):

- Transaction Processing System are information system that processes data resulting from the occurrences of business transactions
- Their objectives are to provide transaction in order to update records and generate reports i.e to perform store keeping function
- The transaction is performed in two ways: **Batching processing** and **Online transaction processing**.
- **Example:** Bill system, payroll system, Stock control system.
- **In** any organization can be multiple transaction processing system like
- For Customer invoice billing system.
- For calculation of raw material production system.

2. Management Information System (MIS):

- Management Information System is designed to take relatively raw data available through a Transaction Processing System and convert them into a summarized and aggregated form for the manager, usually in a report format. It reports tending to be used by middle management and operational supervisors.
- Many different types of report are produced in MIS. Some of the reports are a summary report, on-demand report, ad-hoc reports and an exception report.
- Management information system used to convert the data into information by using tool, algorithms etc.
- MIS is used by the tactical manager for monitoring the current performance of any organization and also help to predict the future performance of organization.
- MIS analysis the data and convert into useful information which is help to support decision making.

- **Example:** Sales management systems, Human resource management system.

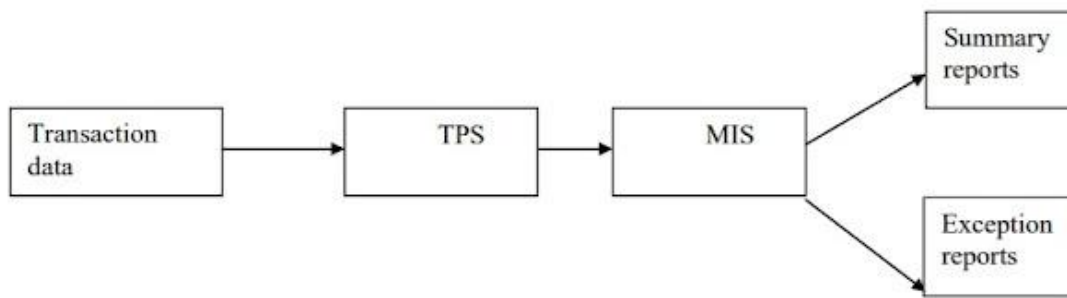


Fig: Interaction between TPS and MIS for Information Needs of an Organisation.

3. Decision Support System (DSS):

- Decision Support System is an interactive information system that provides information, models and data manipulation tools to help in making the decision in a semi-structured and unstructured situation.
- Decision Support System is used by the senior management for taking the decision.
- Decision Support System comprises tools and techniques to help in gathering relevant information and analyze the options and alternatives, the end user is more involved in creating DSS than an MIS.
- **Example:** Financial planning systems, Bank loan management systems.

Components of a DSS

Following are the components of the Decision Support System –

- **Database Management System (DBMS)** – To solve a problem the necessary data may come from internal or external database. In an organization, internal data are generated by a system such as TPS and MIS. External data come from a variety of sources such as newspapers, online data services, databases (financial, marketing, human resources).
- **Model Base Management System** – It take data from data base management system. It stores and accesses models that managers use to make decisions. Such models are used for designing manufacturing facility, analyzing the financial health of an organization, forecasting demand of a product or service, etc.

- **User interface** – Support tools like online help; pulls down menus, user interfaces, graphical analysis, error correction mechanism, facilitates the user interactions with the system.

Types of DSS:

Following are some typical DSSs –

- **Status Inquiry System** – It helps in taking operational, management level, or middle level management decisions, for example daily schedules of jobs to machines or machines to operators.
- **Data Analysis System** – It needs comparative analysis and makes use of formula or an algorithm, for example cash flow analysis, inventory analysis etc.
- **Information Analysis System** – In this system data is analyzed and the information report is generated. For example, sales analysis, accounts receivable systems, market analysis etc.
- **Accounting System** – It keeps track of accounting and finance related information, for example, final account, accounts receivables, accounts payables, etc. that keep track of the major aspects of the business.
- **Model Based System** – Simulation models or optimization models used for decision-making are used infrequently and creates general guidelines for operation or management.

4. Experts System:

- Experts systems include expertise in order to aid managers in diagnosing problems or in problem-solving. These systems are based on the principles of artificial intelligence research.
- Experts Systems is a knowledge-based information system. It uses its knowledge about a specify are to act as an expert consultant to users.

- Knowledgebase and software modules are the components of an expert system. These modules perform inference on the knowledge and offer answers to a user's question
- There are three components of expert system.
- 1. **User interface:** This is the first component of expert system. This component is used to interact with user.
- 2. **Inference engine:-** This is the second component of expert system.
- It is the brain of expert system.
- Its gives the answer of question asked by the users.
- This engine solves the problem on the knowledge base.
- 3. **Knowledge base:-** This is a storage which is used to store the expert knowledge that is used to present the fact and rules.

5. Electronic data processing (EDP)

Electronic data processing is also known as EDP, a frequently used term for automatic information processing.

It uses the computers to collect, manipulate, record, classification and to summarize data. Because computer has capability to storage large amount of data.

EPD can be described as the processing of data using electronic means such as computers, calculators, servers and other similar electronic data processing equipment. A computer is the best example of an **EDP system**.

EDP is the practice of processing, storing, retrieving, sharing and maintaining information electronically.

There are many different ways to process data. The most common way of data processing is the machine learning technique called classification.

Advantage of EDP

Electronic data processing offers various advantages over any other form as it is fast, accurate, reliable and flexible.

E-commerce industry, banking services, record keeping, transaction data processing and any company dealing with real time data processing uses electronic data processing.

Any data processing system comprises 4 basic components, these include hardware, software, procedure & person.

Methods of Electronic Data Processing:

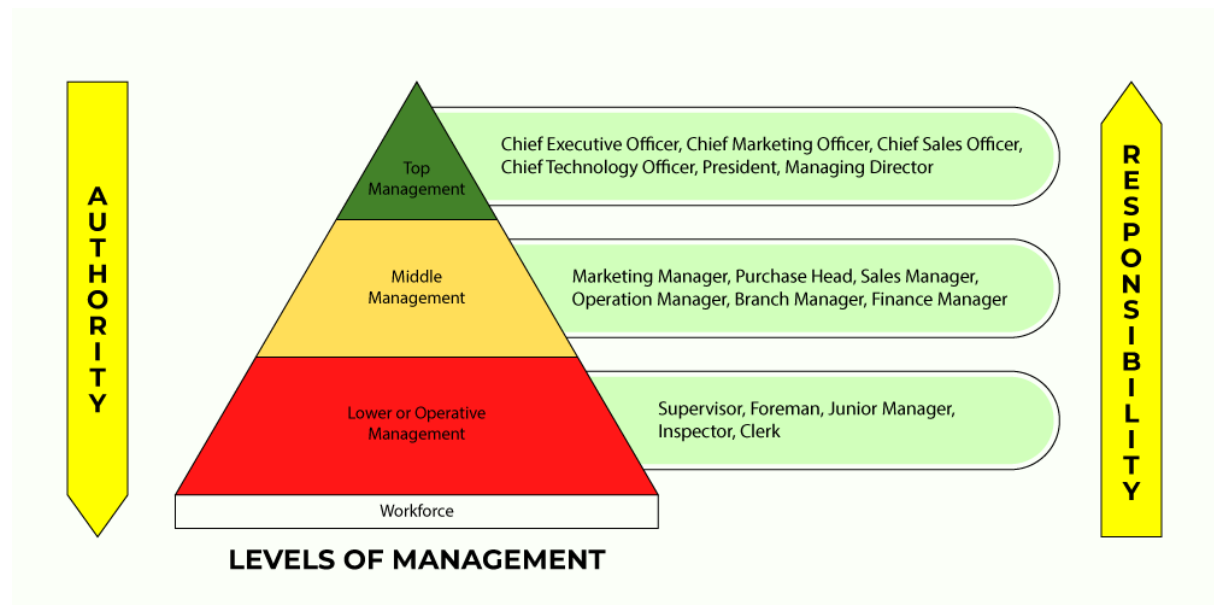
- **Time-sharing:** In this processing method, many nodes connected to a CPU accessed central computer. A multi-user processing system controls the time allocation to each user. Each user can allocate the time slice in a sequence of the Central Processing Unit.
- **Real-time processing:** It Providing accurate and up-to-date information is the primary aim of real-time processing. It is possible when the computer processes the incoming data. It will give the immediate response to what may happen. It would affect the upcoming events. Making a reservation for train and airline seats are the best example for real-time processing.
- **Online processing:** In this processing method, the data is processed instantly. A communication link helps to connect the computer to the data input unit directly. The data input may include a network terminal or online input device. Online processing is mostly used for information research and recording.
- **Multiprocessing:** Multiprocessing is processing of more than one task that uses the different processors at the same time of the same computer. It is possible in network servers and mainframes. In this process, a computer may consist of more than one independent CPU. This makes data processing much faster.
- **Multitasking:** It is an essential feature of data processing. Working with different processors at the same time is called multitasking. In this process, the various tasks share the same processing resource. The operating systems in the multitasking process are time-sharing systems.
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Levels of Management

The term **Levels of Management** refers to the line of division that exists between various managerial positions in an organization. As the size of the company and workforce increases, the number of levels in management increases along with it, and vice versa. The different Levels of Management can determine the chain of command within an organization,

as well as the amount of authority and typically decision-making influence accrued by all managerial positions.

Levels of Management can be generally classified into three principal categories, all of which direct managers to perform different functions.



1. Administrative, Managerial, or Top Level of Management

This level of management consists of an organization's **board of directors and the chief executive or managing director**. It is the ultimate source of power and authority, since it oversees the goals, policies, and procedures of a company. Their main priority is on the strategic planning and execution of the overall business success.

The roles and responsibilities of the top level of management can be summarized as follows:

- Laying down the objectives and broad policies of the business enterprise.
- Issuing necessary instructions for the preparation of department-specific budgets, schedules, procedures, etc.
- Preparing strategic plans and policies for the organization.
- Appointing the executives for middle-level management, i.e. departmental managers.
- Establishing controls of all organizational departments.
- Since it consists of the Board of Directors, the top management level is also responsible for communicating with the outside world and is held accountable towards an organization's shareholders for the performance of the enterprise.

- Providing overall guidance, direction, and encouraging harmony and collaboration.

2. Executive or Middle Level of Management

The **branch and departmental managers** form this middle management level. These people are directly accountable to top management for the functioning of their respective departments, devoting more time to organizational and directional functions. For smaller organizations, there is often only one layer of middle management, but larger enterprises can see senior and junior levels within this middle section.

The roles and responsibilities of the middle level of management can be summarized as follows:

- Executing the plans of the organization in accordance with the policies and directives laid out by the top management level.
- Forming plans for the sub-units of the organization that they supervise.
- Participating in the hiring and training processes of lower-level management.
- Interpreting and explaining the policies from top-level management to lower-level management.
- Sending reports and data to top management in a timely and efficient manner.
- Evaluating the performance of junior managers.
- Inspiring lower level managers towards improving their performance.

3. Supervisory, Operative, or Lower Level of Management

This level of management consists of **supervisors, foremen, section officers, superintendents**, and all other executives whose work must do largely with HR oversight and the direction of operative employees. Simply put, managers at the lower level are primarily concerned with the execution and coordination of day-to-day workflow that ensure completion of projects and that deliverables are met.

The roles and responsibilities of the lower level of management can be summarized as follows:

- Assigning jobs and tasks to various workers.
- Guiding and instructing workers in day-to-day activities.

- Overseeing both the quality and quantity of production.
- Maintaining good relations within lower levels of the organization.
- Acting as mediators by communicating the problems, suggestions, and recommendatory appeals, etc. of workers to the higher level of management, and in turn elucidating higher-level goals and objectives to workers.
- Helping to address and resolve the grievances of workers.
- Supervising and guiding their subordinates.
- Taking part in the hiring and training processes of their workers.
- Arranging the necessary materials, machines, tools, and resources, etc. necessary for accomplishing organizational tasks.
- Preparing periodical reports regarding the performance of the workers.
- Upholding discipline, decorum, and harmony within the workplace.
- Improving the enterprise's image as a whole, due to their direct contact with the workers.

Levels of Management Summary

An organization can have many different managers, across a variety of titles, authority levels, and levels of the management hierarchy that we illustrated above. In order to properly assign roles and responsibilities to all managerial positions, it is important to recognize the key differences between low-level, middle-level, and top-level management.

The key takeaways from this distinction are as follows:

- Top-level managers are responsible for controlling and overseeing the entire organization.
- Middle-level managers are responsible for executing organizational plans which comply with the company's policies. They act as an intermediary between top-level and low-level management.
- Low-level managers focus on the execution of tasks and deliverables, serving as role models for the employees they supervise.

All businesses are comprised of a vast array of different managerial tasks. When these are coordinated properly, and there is a strong hierarchical manager system in place, an

organization can be extremely efficient in creating value through the production of their products, services and overall workflow.
